

Mobility Solution: Four-Wheel Differential Drive on a Single-Pivot Rocker

I. Why Wheels

Europa's surface is fractured, chaotic terrain, sitting at 90 K under 1.314 m/s^2 of gravity. The mission needs the rover to build a spatially continuous subsurface profile along a track. REQ.03.01 calls for a radar trace every 100 m or better along a single transect, and REQ.06.01 requires a sounding stop at each of those intervals. A hopper or a legged walker produces discrete samples, and every landing creates a gap where the localization solution has to start over. Wheels produce a continuous odometry signal, and that continuity is what the whole $\pm 0.5\text{ m}$ localization budget in REQ.02.01 depends on.

Low gravity favors hopping and legged locomotion on paper. A hop costs little energy at 1.3 m/s^2 , and legs manage fractured blocks more gracefully than a rigid wheel does. Both mechanisms hinge on a single impact landing correctly, every time. A hop that lands wrong on a 90 K ice block can end the mission on the spot, and the 44 min one-way delay leaves no window for Earth to step in before the damage is done. A wheeled rocker degrades slowly, bearing by bearing, over many sols. When the recovery loop runs that long, that slow failure mode is more important than the energy a hop would save.

II. Configuration

Four independently driven wheels ride on a single-pivot rocker, with no steering actuators. The chassis is the $0.90\text{ m} \times 0.60\text{ m}$ equipment deck already carried in the mass budget, held 0.35 m above the surface so the belly-mounted sounder keeps its standoff. The wheelbase measures 0.70 m and the track 0.80 m. Four tapered landing struts get jettisoned after checkout, leaving only the wheels in contact with the ice during the survey.

- a. **Wheels.** 0.32 m in diameter, 0.20 m wide, a rigid aluminum rim carrying 24 hardened cleats. The contact path stays bare metal, since anything compliant would degrade fast under a 90 K soak and the thermal cycling that follows.
- b. **Suspension.** One rocker pivot sits on the chassis belly centerline, with bent arms and elbow joints running out to each hub. A rocker-bogie layout, six wheels split across two bogies, would climb better. It also adds two actuators and two thermal isolation paths, on top of a mass budget already sitting at 136.9 kg against a 147.25 kg cap.
- c. **Steering.** Differential drive handles it. Opposing wheel rates rotate the chassis about the rocker pivot, so every heading change is a deliberate skid across the ice.
- d. **Speed.** Commanded speed is 2 cm/s, driven in arcs no longer than 5 m.

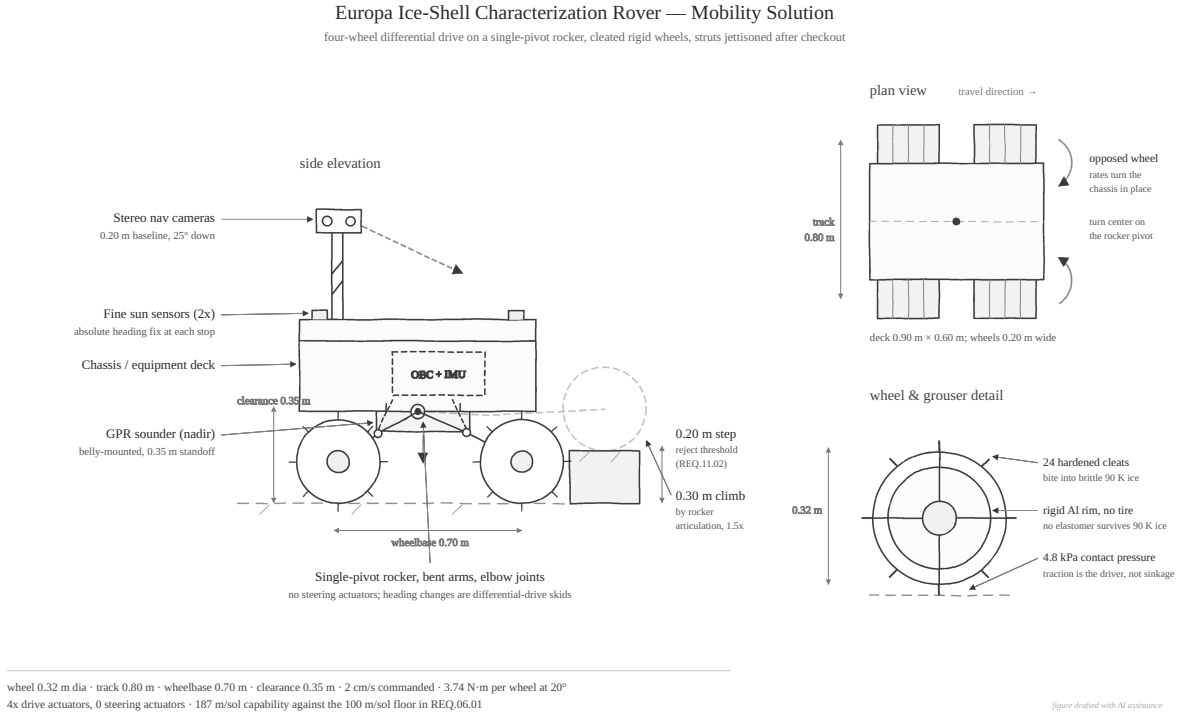


Figure 1: Mobility solution: side elevation, plan view, and wheel detail. *(Figure drafted with AI assistance.)*

III. Traction and Load

Take the mass budget at its contingency cap, 147.25 kg. On Europa that weighs $147.25 \times 1.314 = 193.5 \text{ N}$, 48.4 N per wheel. Spread across a contact patch of about $0.20 \text{ m} \times 0.05 \text{ m}$, that comes to 4.8 kPa. Water ice has a compressive strength in the megapascals, well above that number, so the wheels don't need Mars-style flotation sizing to avoid sinking in. Their job is generating shear.

Tractive force needed to hold the 20° slope limit that REQ.03.02 already imposes, with a rolling resistance coefficient of 0.15:

$$F = W (\sin 20^\circ + 0.15 \cos 20^\circ) = 193.5 \times (0.342 + 0.141) = 93.4 \text{ N}.$$

That force works out to 23.4 N per wheel, 3.74 N·m at a 0.16 m radius. At 2 cm/s the useful mechanical power comes to just 1.87 W. Nearly all of the 84 W peak already budgeted for the four drive motors covers breakout torque and gearbox drag at 90 K, where stiff lubricant determines the load; moving the rover itself barely registers against that number. REQ.09.02 preheats the actuators above -40°C before any commanded motion for the same reason.

The available shear favors this design, for a reason specific to Europa. Ice turns slippery near its melting point because contact pressure forms a thin meltwater film. At 90 K that film doesn't form, and the friction coefficient for a hard slider on ice climbs toward 0.5 to 0.7, far above the 0.05 you'd measure on a temperate glacier [1]. Using 0.5, the limiting slope comes to 26.6° against the required 20° , so the cleats carry margin without needing an aggressive grouser height. Europa's ice

behaves like brittle rock at this temperature, which helps traction and wears at the cleats, so the cleats are separate hardened inserts bolted onto the rim.

Per-sol distance comes in well above the requirement. Four hours of drive time at 2 cm/s, with a 65% duty cycle after subtracting stops for imaging and hazard assessment, works out to 187 m against the 100 m floor in REQ.06.01. REQ.06.01 only demands an average rate of 6.9 mm/s, so hazard rejection and the sounding cadence set the pace of the traverse, before the drive train ever does.

IV. Steering Without Steering Actuators

Skid steering slips badly on hard ice. A differential turn on a surface with $\mu \approx 0.5$, with no compliance anywhere in the contact patch, produces wheel rotation that doesn't match chassis rotation, so wheel odometry can't close a turn on its own. The design routes around that by dropping wheel odometry as a position source and using it as a slip observable instead: the residual between encoder counts and stereo visual odometry becomes the slip estimate, the same approach the MER rovers used on Martian sand [2].

That approach works because heading gets re-anchored with an absolute fix at regular intervals, so error never gets the chance to integrate across the whole traverse. A tactical-grade MEMS IMU holding $1.5^\circ/\text{h}$ of bias stability drifts 1.04° across a full 50 m segment at 2 cm/s. That alone comes to 0.91 m of cross-track error, twice the ± 0.5 m budget in REQ.02.01. Breaking the segment into 5 m arcs and taking a sun-sensor heading fix at the end of each one caps the inertial contribution at 0.104° . Combined with the $\pm 0.25^\circ$ fix accuracy, the heading error lands at 0.271° , which over 50 m comes to 0.24 m of cross-track error. The full budget closes out like this:

Error source over a 50 m segment	Contribution (m)
Visual odometry scale and feature residual	0.30
Heading knowledge (0.271°)	0.24
Wheel slip on cleated water ice	0.20
Inter-agent range residual	0.15
Root-sum-square	0.46
REQ.02.01 limit	0.50

Table 1: Localization error budget for one 50 m traverse segment.

Eight percent of margin is thin. If slip comes in worse than assumed, it's the term to attack first, since a mobility change can move it while the other terms depend on sensors already picked. Widening the wheels or adding grouser height buys shear directly. Swapping the differential turn for four steering actuators would buy more margin, at a cost of roughly 6 kg and four more cold-start mechanisms, so that trade stays on the shelf unless analog testing pushes the slip term above 0.30 m.

V. Alignment With the Mission

This mobility design answers commitments the mission already made. The single rocker pivot holds $\pm 0.5^\circ$ of tilt knowledge, which keeps the sounder within $\pm 5^\circ$ of nadir on uneven ice, ten

times the margin REQ.03.02 needs. Localization error comes out to 0.46 m per segment, inside the REQ.02.01 budget, and that's what lets the continuous, georeferenced along-track profile in REQ.03.01 actually mean something. Per-sol capability runs to 187 m against a 100 m requirement, so the schedule has room to absorb whatever navigation holds REQ.11.04 forces along the way.

The rocker fails in a forgiving way. Lose one drive actuator, and three wheels can still drag the chassis through a reduced-rate traverse. A four-corner steered design doesn't get that grace: lose one steering actuator there and a wheel locks at an angle, fighting every arc the rover tries afterward. Nobody is flying out to fix either failure, so the independent failure mode matters more than shaving weight or squeezing out nominal performance.

VI. Open Items

A few things are still open. Cleat wear against brittle ice over a 30-sol traverse has no flight precedent, and abrasion at 90 K could shift the effective friction coefficient in either direction. The 0.15 rolling resistance coefficient comes from Martian regolith rather than measured ice analog data; treat the 3.74 N m torque number as an estimate until a bench test replaces it. The mass and power budgets don't carry mobility line items yet either; that gap is flagged separately, along with the 17.1 kg and 3.5 W this design adds.

References:

References

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AI disclosure: I used AI to check the arithmetic behind the traction, torque, and localization error budget numbers, and to cross-check the low-temperature ice friction values against published measurements. The mobility architecture is all mine. Diagram generated using Claude Opus 5.